**shock front.** A place, in flowing gas, where the density and temperature of the gas suddenly jump upward by a large amount.

**simultaneity breakdown.** The fact that events which are simultaneous as measured in one reference frame are not simultaneous as measured in another frame that moves relative to the first.

**singularity.** A region of spacetime where spacetime curvature becomes so strong that the general relativistic laws break down and the laws of quantum gravity take over. If one tries to describe a singularity using general relativity alone, one finds (incorrectly) that tidal gravity and spacetime curvature are infinitely strong there. Quantum gravity probably replaces these infinities by quantum foam.

**Sirius B.** The white-dwarf star that orbits around the star Sirius.

**spacetime.** The four-dimensional "fabric" that results when space and time are unified.

**spacetime curvature.** The property of spacetime that causes freely falling particles that are initially moving along parallel world lines to subsequently move together or apart. Spacetime curvature and *tidal gravity* are different names for the same thing.

**spacetime diagram.** A diagram with time plotted upward and space plotted horizontally.

**special relativity.** Einstein's laws of physics in the absence of gravity.

**spectral lines.** Sharp features in the. spectrum of the light emitted by some source. These features are due to strong emission or absorption at specific wavelengths, produced by specific atoms or molecules.

**spectograph.** A sophisticated version of a prism, for separating the various colors (wavelengths) of light and thereby measuring the light's spectrum.

**spectrum.** The range of wavelengths or frequencies over which electromagnetic waves can exist, running from extremely low-frequency radio waves up through light to extremely high-frequency gamma rays; see Figure P.2 in the prologue. Also, a picture of the distribution of light as a function of frequency (or wavelength), obtained by sending the light through a prism.

**spin.** Rotation. See *angular momentum*

**stability.** The issue of whether an object is unstable or not. See also *unstable.*